Radar Human Activity Detection

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CONCEPT OF OPERATIONS

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FOR
Human Activity Radar Detector

TEAM 4

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Table of Contents

Table of Contents ..... **IV**
List of Tables ..... **V**
No table of figures entries found. ..... **V**
List of Figures ..... **VI**
Figure 1: Human Radar Detector Block Diagram ..... **VI**
1. Executive Summary **1**
2. Introduction **1**
2.1. Background 1
2.2. Overview 2
2.3. Referenced Documents and Standards 2
3. Operating Concept **3**
3.1. Scope 3
3.2. Operational Description and Constraints 3
3.3. System Description 3
3.4. Modes of Operations 4
3.5. Users 4
3.6. Support 4
4. Scenario(s) **4**
4.1. Assisted Living Fall Detection 4
4.2. Gym Fall Detection 5
5. Analysis **5**
5.1. Summary of Proposed Improvements 5
5.2. Disadvantages and Limitations 5
5.3. Alternatives 5
5.4. Impact 6

List of Tables

No table of figures entries found.

List of Figures

Figure 1: Human Radar Detector Block Diagram

1. Executive Summary

Tracking the activities of an individual in an assisted-living home is essential in ensuring their continued physical well-being. However, doing so often requires means that unintentionally violate the privacy rights of these individuals, such as the installation of equipment that collects visual, audio, and other forms of data. These are usually installed in private areas in which the individual cannot receive assistance in a timely manner. As such, it is critical that a product be developed that both reliably detects & alerts the proper personnel for emergency assistance and does not intrude on individual privacy. This product is designed to collect data using radar signal reflections to assess the received signal. Using a machine learning model trained on collected data sets of falling individuals, the fall signal will be continuously classified over a period of ten to twenty seconds. If the pattern persists, an alert will be transmitted to a mobile application interface for assisted-living personnel so that further emergency action can be taken. The scope of this project is to provide a prototype that increases the efficiency of response times by assisted-care workers at a minimal level of personal privacy violation.

2. Introduction

This document introduces the human activity radar detector, which monitors private areas in assisted-living homes for signs of fall injury in order to alert on-site personnel for further emergency care. The module will reduce the frequency and time between injury and the arrival of assistance and will ensure that the privacy of vulnerable individuals in these spaces is not violated.

2.1. Background

Injury from falling is common among the elderly in assisted-living homes, and such injuries can cause further physical ailments if the fall goes unnoticed. According to a collection of 18 studies on elderly individuals living in nursing homes, the incidence of falls is 43% [1]. This means that a significant minority of the elderly in these facilities have experienced some form of injury from falling. Therefore, it is critical to develop a product that minimizes the potential physical impairments that result from these injuries.

The protection of individual privacy is another key aim of this project. In recent times, the usage of camera systems for home security has led to privacy failures and breaches of trust on the part of service providers. For example, in 2023, the home security company Ring did not invest in adequate privacy & security provisions of customer data, which led to the misuse of the data by employees [2]. This presents a serious risk for users of home cameras, especially for those in assisted care who are more vulnerable, and a need for an alternative that can accomplish the same tasks without violating individual privacy. This project aims to develop a system that utilizes minimally invasive data collection, which will ensure the security of users' privacy.

2.2. Overview

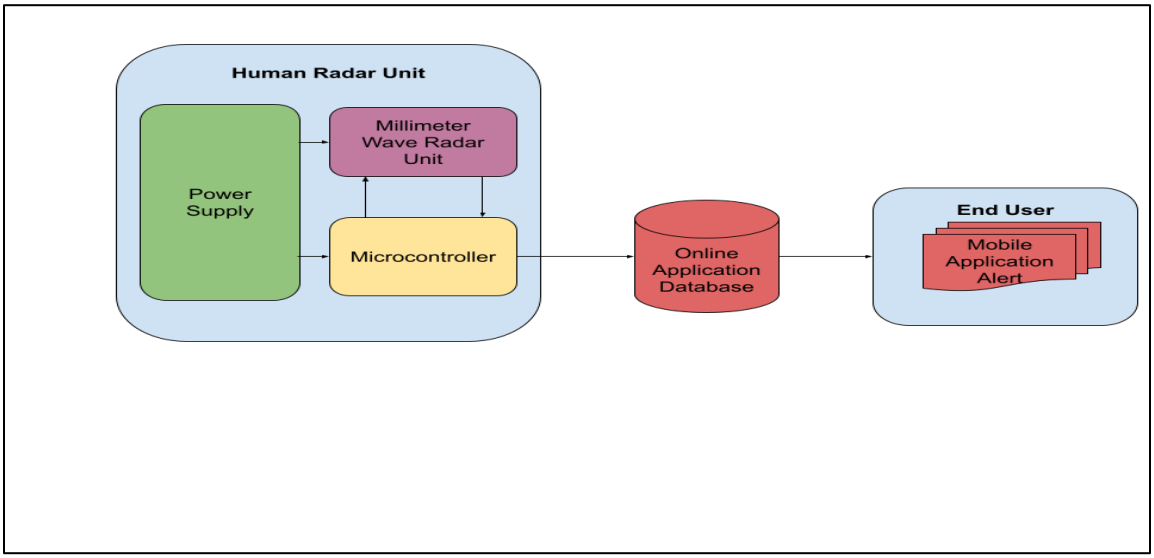

Figure 1: Human Activity Radar Detector Block Diagram

This system will be used to monitor a set location for people falling. This will be done by three primary components within the Human Radar Unit: the power supply, the microcontroller, and a mobile app interface to display the location of the received signal. The human activity radar detector will be outfitted with a radar module, which includes transmit and receive antennas, to detect objects at a range of eight to ten meters. This radar module will continually transmit & receive signals, which, using an onboard signal processing module, will send digital data to a microcontroller. The microcontroller will be programmed with a machine learning algorithm that will be trained on data sets of fall injury radar patterns. By classifying the received signal, the module will alert the on-site personnel through a mobile app interface so that they can provide the necessary care.

2.3. Referenced Documents and Standards

[1] Shao L, Shi Y, Xie XY, Wang Z, Wang ZA, Zhang JE. Incidence and Risk Factors of Falls Among Older People in Nursing Homes: Systematic Review and Meta-Analysis. J Am Med Dir Assoc. 2023 Nov;24(11):1708-1717. doi: 10.1016/j.jamda.2023.06.002. Epub 2023 Jul 8. PMID: 37433427.

[2] Staff, BCP, et al. "Ring's Privacy Failures Led to Spying and Harassment through Home Security Cameras." *Consumer Advice*, 30 Oct. 2023, consumer.ftc.gov/consumer-alerts/2023/05/rings-privacy-failures-led-spying-harassment-through-home-security-cameras.

3. Operating Concept

3.1. Scope

The Human Activity Radar Detector (HARD) will be used in residential and assisted living settings to detect human falls within a short distance (below 10m) using a high-frequency radar sensor. The radar sensor mounted on the wall will constantly be sending data to a control unit, where a microcontroller running a classification machine learning model will determine if there has been a fall injury. Once a fall injury has been detected, the microcontroller will communicate with an app via Wi-Fi to send out an alert to caretakers/family and emergency services.

3.2. Operational Description and Constraints

The Human Activity Radar Detector will be installed in spaces where assisted care is at the lowest level, such as bathrooms, bedrooms, etc. The module will transmit & receive signal reflections that will be processed to determine the occurrence of a falling injury, verify the certainty that the injury has in fact occurred, and alert the nearby assisted-care worker through an app interface. Below are listed the constraints of this design:

- The app interface must be able to identify the designated room location of a signal transmission in case of emergency.
- The module must be able to operate over the Wi-Fi network or otherwise be able to access an internet connection.
- The machine learning algorithm for detecting signal patterns will be trained on a predetermined set of data, which cannot be modified in the future.
- The system must be able to operate using a 9V battery that will be regulated by on-board circuitry.
- The system must continually classify affirmative signals for a period of 10-20 seconds before alerting personnel to prevent false alarms from triggering.

3.3. System Description

The radar detection system will be defined by three central parts: the radar detector, the microcontroller programmed with a machine learning algorithm, and the mobile application user interface. Power will be provided by a peripheral power management system. The radar detector will deliver digitally processed FMCW data to the microcontroller, which will classify the signal using a machine learning algorithm. The algorithm will decide whether the signal resembles a fall injury. If so, a signal will be transmitted over Wi-Fi to the user interface to indicate that a fall has occurred.

3.4. Modes of Operations

The system will have one primary mode of operation, in which detected falls are tracked with an alert sent to a mobile device. This will consist of three primary actions performed by the system: detection, validation, and finally alerting the user.

Detection
The system will emit & process signals continuously to identify whether an injury pattern has occurred.

Validation
The system will continue to process signals as before; however, the system will also maintain a counter that will verify that an injured person has remained on the floor after the fall.

Alert
The system will send an alert to the user interface to inform assisted-care personnel in the area that an injury has occurred.

3.5. Users

This system will primarily be used by individuals who are at risk of serious injury from falls and their immediate family or caretakers. The level of training required to use this system will be very low, with the only requirements being the installation of a mobile app and the mounting of the system on a wall. This system will benefit both at-risk individuals and their families, providing both of those groups the peace of mind that if something bad were to happen, they would be informed of it in a timely manner along with the appropriate medical personnel.

3.6. Support

This system will include a user manual in the box detailing how to install the physical device and the phone app. If users are having issues with the app, they will be able to either call a support number with their issue or send an email relaying the issue to the manufacturer.

4. Scenario(s)

4.1. Assisted Living Fall Detection

The Human Activity Radar Detector's primary use is for fall detection in assisted living environments. Residents in assisted living are at high risk of falls, which, if left unattended, can result in compounded side effects. The proposed radar system would provide continuous, non-invasive monitoring of desired areas in residential care homes (possibly in bedrooms/bathrooms where staff are less present). When falls are detected, the system will alert the staff via a phone/web app, ensuring a rapid response.

4.2. Gym Fall Detection

Another possible application of this system is to work within public gyms. Given the irregular hours that gyms can be open, having personnel on site constantly monitoring issues could be difficult. Areas or exercises where fall detection could be applicable include the squat rack, deadlift, power clean, and other such exercises with a full range of motion. Having an automated alert system could help employees watch for possible life-threatening issues while reducing mental workload

4.3. Child/Animal in Car Detection

Another application of radar detection could be detecting children or animals inside a locked car. There have been numerous cases globally of child or animal deaths due to being stuck in locked cars on a hot day. This issue is prevalent enough that, in the coming years, the EU will require cars to have radar detection systems inside for this exact purpose. There is no practical way for any police body to monitor all cars to protect against this issue. Having a system to check for this can ensure more child and animal lives are saved each year.

5. Analysis

5.1. Summary of Proposed Improvements

- The system will operate off battery power for its main configuration; this will allow placement in locations where a power source is not available.
- Ease of installation: Given that the system's logic is based on a machine learning model, user input will be minimal during both installation and operation.
- Cybersecurity integration: In order to prevent wireless communication that may retrieve user data, integrating basic cybersecurity into the system would greatly advance the stated purpose of protecting users' privacy.

5.2. Disadvantages and Limitations

- Need to ensure that batteries are adequately charged.
- Low computing capacity within the microcontroller limits ML implementation.
- Limited directionality compared to a camera-operated system.
- High cost for FMCW radar sensors.

5.3. Alternatives

Alternatives to the millimeter wave radar detection solution include, but are not limited to:

- Constant monitoring by attending caretakers
- Off-the-shelf camera monitoring systems
- The same unit, but with a wired power solution
- Life-Alert style button
- Smart watch fall detection alerts

5.4. Impact

This design is largely focused on reducing the impact of data collection on personal privacy; however, there remains a minimal level of data that must be collected from the operating environment in order for the design to function. This does not pose any significant risk to the user's personal information or privacy. In addition, the programmed algorithm does not feature the ability to receive updates, modifications, or any devices that record data, which is not explicitly stated in this document, making the user's data secure from malpractice.